

DUEBENDORF AP, Switzerland

WMO: 068070

Lat: 47.402N

Lon: 8.634E

Elev: 1427

StdP: 13.95

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.3	21.5	12.1	10.7	18.4	17.3	13.9	24.7	20.2	48.3	16.9	44.9	3.7	120	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	88.1	68.5	84.5	67.3	81.0	66.0	70.4	84.3	68.4	80.8	67.2	78.8	5.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	100.7	76.4	64.2	94.9	74.4	62.7	89.9	72.3	35.1	84.8	33.7	81.7	32.5	78.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.6	11.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	10.1	73.7	6.0	1.7	5.8	75.0	2.4	75.9	-1.0	76.9	-5.3	78.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	35.6	34.7	42.5	51.2	57.5	65.4	68.4	66.6	59.9	51.8	42.6	36.7
	DBStd	13.64	7.30	7.73	6.93	7.00	6.74	6.31	5.82	5.95	5.20	6.45	6.40	6.32
	HDD50	1912	447	429	245	70	13	1	0	0	1	60	234	412
	HDD65	5402	910	849	699	415	246	70	33	52	169	410	672	877
	CDD50	2337	2	0	11	105	246	464	569	515	297	115	12	1
	CDD65	353	0	0	0	0	14	83	138	102	15	1	0	0
	CDH74	3316	0	0	0	36	187	831	1256	881	122	3	0	0
	CDH80	1034	0	0	0	2	34	264	438	283	13	0	0	0
(14) Wind	WSAvg	3.6	4.2	3.6	4.3	3.7	3.9	3.7	3.7	3.2	3.1	2.9	3.2	3.6
(15) Precipitation	PrecAvg	47.5	2.7	2.8	3.0	3.5	5.2	5.4	5.5	5.2	3.8	3.5	3.5	3.5
	PrecMax	60.7	6.7	6.4	8.9	6.9	9.6	8.7	10.1	9.3	8.2	6.3	7.7	7.3
	PrecMin	37.4	0.4	0.8	0.9	0.5	1.2	2.0	2.1	0.7	1.8	0.3	0.0	0.0
	PrecStd	6.1	1.7	1.5	1.7	1.7	2.3	1.7	2.1	2.0	1.6	1.6	1.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	58.7	67.8	77.4	84.0	91.8	93.2	91.2	81.0	72.0	63.0	55.3
		MCWB	50.5	46.6	52.1	57.9	64.4	71.5	70.0	69.3	66.5	60.0	53.9	48.2
	2%	DB	53.4	53.7	62.8	73.6	78.9	86.3	89.2	86.4	77.3	68.3	59.2	52.0
		MCWB	47.2	44.8	49.8	56.3	62.3	67.6	68.4	68.8	64.6	59.2	52.6	47.4
	5%	DB	49.7	49.8	58.9	70.0	75.1	82.5	84.6	82.7	73.5	64.8	55.5	49.7
		MCWB	45.0	43.2	48.4	54.9	60.4	66.6	67.1	67.9	62.3	57.7	50.0	45.2
	10%	DB	45.0	45.0	55.1	66.1	71.3	78.6	80.9	78.8	70.0	61.2	52.0	46.2
		MCWB	41.9	40.5	46.5	53.1	58.1	64.9	65.7	66.6	61.2	56.0	48.2	43.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	48.2	53.0	59.5	65.5	73.5	71.9	72.0	67.8	62.1	55.5	50.0
		MCDB	56.1	57.8	65.1	74.3	80.2	88.8	87.3	84.5	78.3	69.6	62.6	53.1
	2%	WB	47.9	45.5	50.6	57.5	63.4	69.9	69.9	70.1	65.6	60.2	52.7	47.9
		MCDB	52.5	52.4	60.9	71.1	76.2	83.5	84.7	83.4	75.6	66.3	58.4	51.7
	5%	WB	44.9	43.2	48.9	55.5	61.3	67.5	68.2	68.3	63.7	58.5	50.7	45.6
		MCDB	48.6	48.6	57.4	68.7	71.7	79.7	80.6	80.0	72.1	63.9	55.3	49.2
	10%	WB	42.1	40.9	46.9	53.6	59.3	65.6	66.9	67.0	62.0	56.8	48.8	42.9
		MCDB	44.6	44.8	54.1	64.6	68.8	76.7	78.9	77.7	69.0	61.0	51.9	45.1
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	12.9	17.0	19.4	18.5	19.3	18.9	18.2	17.1	14.7	11.4	10.5
		MCDBR	14.8	22.1	25.2	28.2	26.1	26.7	26.3	25.2	22.4	20.7	17.5	15.5
		MCWBR	10.2	14.2	14.5	14.1	12.0	11.2	9.7	10.5	11.3	12.3	11.9	10.9
	5% WB	MCDBR	14.2	20.1	22.9	25.9	23.1	24.1	23.3	22.8	20.2	19.0	16.8	14.5
		MCWBR	10.2	13.3	13.5	13.3	10.9	10.5	9.5	10.0	10.6	11.9	11.4	10.5
(40) Clear-Sky Solar Irradiance	taub		0.292	0.317	0.353	0.389	0.403	0.408	0.411	0.403	0.376	0.349	0.317	0.288
	taud		2.441	2.373	2.335	2.271	2.276	2.304	2.292	2.329	2.384	2.448	2.483	2.452
	Ebn at Noon		253	268	274	274	273	272	269	266	263	253	241	240
	Edh at Noon		21	28	34	40	41	40	40	37	32	26	20	19
(44) All-Sky Solar Radiation	RadAvg		375	644	1027	1376	1579	1814	1756	1533	1151	726	407	308
	RadStd		41	86	138	199	159	168	160	157	118	79	50	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (117 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.28	+0.76	+0.61	N/A	N/A	+103	+29

Nomenclature: See separate page